

Madhav Gupta

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EDUCATION

Stony Brook University

B.S. in Computer Science

Stony Brook, New York

Expected Graduation, May 2028

- **GPA:** 3.66/4.00, *Dean's List*
- **Related Coursework:** Data Structures & Algorithms, Objects & Design, Computer Organization & Programming, Compiler Design, Database Systems, Object-Oriented Programming, Statistics & Applications

SKILLS

- Programming Languages & Databases: Python, Java, JavaScript (Node.js), C, SQL, MongoDB, PostgreSQL.
- DevOps & Cloud Security: GitHub Actions (CI/CD), Docker (Multi-stage), Terraform (IaC), AWS (VPC/EKS), Linux (Ubuntu/Kali).
- Security Tools & Analysis: Splunk Enterprise, Wireshark, Trivy (SCA), Checkov, SIEM Log Analysis, MITRE ATT&CK, UTM.
- Web & Frameworks: React.js, Node.js, RESTful APIs, Microservices.

EXPERIENCE

Manras Technologies

Frontend Development Intern

Zirakpur, India

June 2025 – Aug 2025

- Provided end-to-end technical support, completing **100%** of documentation, software updates, and troubleshooting requests within deadlines.
- Optimized Dynamics data models by refining record types, page layouts, and validation rules, reducing form errors by **20%** and improving system reliability.
- Created and managed **60+ high-priority** Visualforce and PDF documents for Adani Group real estate projects, ensuring accurate delivery of client-facing materials.
- Collaborated with team members on document assignments, streamlining handoffs and improving task completion efficiency across the portfolio.

Stony Brook University – AMS 161: Calculus II

Teaching Assistant

Stony Brook, New York

August 2025 – December 2025

- Assisted professor in facilitating coursework for **200+ undergraduate students**, reinforcing concepts in integration techniques, sequences, and series.
- Led weekly recitation sessions and office hours, clarifying problem-solving strategies and boosting student comprehension.
- Graded homework and exams with **100% accuracy** and timely feedback, ensuring fairness and consistency.

PROJECTS

SOC Detection Lab

Stony Brook, New York

- Designed a local security operations environment utilizing UTM virtualization to safely isolate attacker (Kali Linux) and victim (Ubuntu) network traffic.
- Deployed a Splunk Enterprise SIEM data pipeline, configuring Universal Forwarders to parse and index local system authentication logs (/var/log/auth.log) from a headless server.
- Conducted simulated network reconnaissance to capture .pcap telemetry in Wireshark, mapping the offensive behavior to the MITRE ATT&CK framework.

End-to-End DevSecOps Cloud Pipeline

Stony Brook, New York

- Provisioned secure AWS infrastructure (VPC, EKS) using Terraform and integrated Checkov to automatically scan IaC manifests, preventing cloud misconfigurations prior to deployment.
- Containerized a 3-tier web application (React, Node.js, PostgreSQL) utilizing multi-stage Docker builds and enforced non-root execution privileges to minimize the container attack surface.
- Engineered an automated CI/CD pipeline via GitHub Actions, integrating Trivy and npm audit for Software Composition Analysis (SCA) to proactively block critical CVEs from reaching production.

Astra

Stony Brook, New York

- Built a full end-to-end interpreter for a custom block-structured, ML-style language using Python and PLY (Lex/Yacc), covering lexing, parsing, AST construction, and execution.
- Designed a grammar-driven parser that converts source code into an Abstract Syntax Tree (AST) and enforces syntactic correctness via clear SYNTAX ERROR handling.
- Implemented AST-based semantic evaluation, embedding execution logic within node classes to support expressions, control flow, functions, and data structures.

FormFlow | Runner-Up (MongoDB Track), Code a Site Hackathon 2026

Stony Brook, New York

- Engineered a real-time, browser-based AI personal trainer utilizing a 17-point MediaPipe joint mesh to process raw webcam video into actionable biomechanical coaching, tracking joint telemetry and movement tempo at 60fps.
- Secured Runner-Up in the MongoDB track by developing a gamified backend that tracks rep-by-rep movement fluidity and instantly broadcasts form faults (e.g., knee valgus) to synchronized peer groups for gym accountability.